SEX-LINKED INHERITANCE

- This is a form of inheritance where the gene/allele for the characteristic being studied is on the _____ chromosome.
- Diseases such as **colour-blindness** and **haemophilia** are inherited this way, and are more common in _____ than females.

HAEMOPHILIA

- **Alleles, Genotypes and Phenotypes for Haemophilia (Blood-Clotting Inability)**

  X_H – allele for normal blood clotting
  X_h – allele for haemophilia

  X_HX_H – genotype of normal female
  X_hX_h – genotype of haemophiliac female
  X_HX_h – genotype of carrier female (with normal blood-clotting ability, but who can pass the _____ gene to her children)

  X_HY – genotype of normal male
  X_hY – genotype of haemophiliac male

- **Example 1 – Haemophilia**

  A man with normal blood-clotting ability (X_HY) marries a woman who is a haemophiliac (X_hX_h). **The possible phenotypes of their children are ...**

	X_H	Y
X_h	X_HX_h	X_hY
X_h	X_HX_h	X_hY

  Possible genotypes = 1 X_HX_h : 1 X_hY
  Possible phenotypes
  = 1 _____ female : 1 _____ male

- **Example 2 – Haemophilia**

  A normal woman (X_HX_H) marries a haemophiliac man (X_hY). **The possible genotypes and phenotypes of the children are...**

	X_H	X_H
X_h	X_HX_h	X_HX_h
Y	X_HY	X_HY

  Possible genotypes = 1 X_HX_h : 1X_HY

Possible phenotypes
= 1 _____ female : 1 _____ male

RED-GREEN COLOUR BLINDNESS

- **Alleles, Genotypes and Phenotypes for Colour-blindness**

Xc - allele for normal colour vision
X_c - allele for colour-blindness

Xc Xc - genotype of normal female
X_c X_c - genotype of colourblind female
Xc X_c - genotype of carrier female (with normal colour vision, but who can pass the defective gene to her children)
Xc Y - genotype of normal male
X_c Y - genotype of colourblind male

- **Example 3 – Colour Blindness**

A male with normal vision (XcY) and a colourblind female (X_c X_c) have children. The possible genotypes and phenotypes of the children are ...

	Xc	Y
X_c	XcX_c	X_cY
X_c	XcX_c	X_cY

Possible genotypes = _____

Possible phenotypes
= _____

- **Example 4 – Colour Blindness**

A carrier female (Xc X_c) marries a normal-visioned male (Xc Y). The possible genotypes and phenotypes of the children are...

	Xc	X_c
Xc	XcXc	XcX_c
Y	XcY	X_cY

Possible genotypes
= _____

Possible phenotypes
= _____